$$u_1 \sim U(0,1)$$
$$u_2 \sim U(0,1)$$
$\rightarrow$ مستقل

$$R = \sqrt{-2\ln(u_1)} \rightarrow \bigstar$$

$$\Theta = 2\pi u_2 \longrightarrow f_\Theta(\theta) = \frac{1}{2\pi} \quad 0<\theta<2\pi \longrightarrow$$ می‌دانیم

$$\bigstar: \Pr(R \le r) = \Pr\left(\sqrt{-2\ln(u_1)} \le r\right) = \Pr\left(-2\ln(u_1) \le r^2\right) = \Pr\left(\ln(u_1) \ge -\frac{r^2}{2}\right)$$

$$= \Pr\left(u_1 \ge e^{-\frac{r^2}{2}}\right) = 1 - \underbrace{\Pr\left(u_1 \le e^{-\frac{r^2}{2}}\right)}_{\text{CDF uniform}} = 1 - e^{-\frac{r^2}{2}}$$

$$\boxed{0 < e^{-\frac{r^2}{2}} < 1}$$

$$\rightarrow F_R(r) = 1 - e^{-\frac{r^2}{2}} \longrightarrow f_R(r) = \frac{dF_R(r)}{dr} = re^{-\frac{r^2}{2}} \checkmark$$

و می‌دانیم $u_1 \perp u_2 \longrightarrow R(u_1) \perp \Theta(u_2)$

$$\rightarrow f_{R,\Theta}(r,\theta) = f_R(r)\, f_\Theta(\theta) = \frac{r}{2\pi} e^{-\frac{r^2}{2}}$$

حال تغییر متغیر می‌دهیم:

$$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases} \quad |J| = r \longrightarrow x \perp y$$

$$f_{XY} = f_{R\Theta}(r,\theta) \times \frac{1}{|J|} = \frac{1}{2\pi} e^{-r^2/2} \rightarrow f_{XY} = \frac{1}{2\pi} e^{-\frac{x^2+y^2}{2}}$$

$$f_{XY} = \frac{1}{\sqrt{2\pi}\sqrt{\sigma^2}} e^{-\frac{x^2+y^2}{2\sigma^2}} \longrightarrow \sigma^2 = 1$$

$$f_{XY} = f_X f_Y = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \times \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} \rightarrow f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

دقیقاً همین را می‌خواهیم.

$$\rightarrow x = r\cos\theta \rightarrow x = \sqrt{-2\ln(u_1)}\cos(2\pi u_2) \checkmark$$